

MSIT 5910-01 Capstone Project — Unit 7

Part A: System Testing and Evaluation — Extracted Tables

University of the People | Term 4, AY2026

A1. System Testing

Table 1. System Test Suite — 10 Test Cases Across Integration, System, and Acceptance Levels

TC#	Level	Module Tested	Input	Expected Output	Result
TC-01	Integration	Google Analytics → KPI Calculator	GA sessions data, last 7 days	Sessions metric populates scorecard	PASS
TC-02	Integration	Facebook Insights → Dashboard UI	Live Page post engagements + Total reach	Engagement Rate (%) scorecard populated	PASS
TC-03	Integration	Data Filter → Aggregation Engine	Raw API response including page_id field	page_id absent from processed output	PASS
TC-04	System	Full Dashboard — Date Range Filter	Switch from Last 7 to Last 30 days	All KPIs and charts refresh within 5s	PASS
TC-05	System	Full Dashboard — Post Type Filter	Select 'video' in Post Type dropdown	All Facebook charts show video data only	PASS
TC-06	System	Geo Map rendering	GA data with multi-country traffic	World map shaded by session volume	PASS
TC-07	System	PDF Report Export	Click Share → Download as PDF	Formatted PDF downloaded within 10s	PASS
TC-08	System	Access Control (RBAC)	Non-whitelisted email attempts access	Access denied, redirected to login	PASS
TC-09	Acceptance	Editorial Manager UAT	15-min guided dashboard walkthrough	SUS score ≥ 70, positive feedback	PASS
TC-10	Acceptance	Sales Team UAT	Sales rep reviews exported PDF report	Report usable for advertiser presentation	PASS

Colour key — Level column: Blue = Integration | Amber = System | Green = Acceptance · All 10 test cases returned PASS.

Level	Scope	Cases
Integration	Verifies data flows between API sources, filter, and KPI Calculator without field loss	TC-01 – TC-03
System	Validates fully assembled dashboard against all five functional requirements	TC-04 – TC-08
Acceptance	Confirms real-world usability via guided UAT and SUS questionnaire (score: 82 / Excellent)	TC-09 – TC-10

A2. Test Documentation — Issue Log

Table 2. Issue Log — Three Bugs Resolved Before v1.0.0 Release; Two Limitations Documented

ID	Description	Severity	Resolution	Status
BUG-01	Division-by-zero in Reach Growth formula when prior week reach = 0	High	Added CASE WHEN zero-guard to formula	✓ Resolved
BUG-02	Meta API field names inconsistent with connector documentation	Medium	Field alias normalization added at connector layer	✓ Resolved
BUG-03	Date Range Control not updating Geo Map in View mode	Medium	Set Geo Chart default date range to Auto	✓ Resolved
LIM-01	Broadcast data requires manual CSV upload — not automated	Low	Documented; Make.com automation planned for v1.1	Open
LIM-02	SUS sample of 3 stakeholders is below recommended minimum of 5	Low	Noted as evaluation limitation; expanded testing post-deployment	Open

Severity colour key: Red = High | Amber = Medium | Blue = Low · Status: Green = Resolved | Amber = Open

A3. Evaluation and Reflection Summary

Table 3. Testing Outcomes and Evaluation Contributions

Contribution	Detail	Concept (Summers, 2020)
Bug prevention before production	Caught 3 defects (BUG-01–03) before deployment; BUG-01 would have silently corrupted Reach Growth KPI	Verification — confirming the product is built right
Functional requirements verified	All 5 functional requirements met: KPI display, date/platform filters, PDF export, access control	Verification — all requirements traceable to passing test cases
Real-world usability confirmed	SUS score of 82 (Excellent) achieved in acceptance testing with editorial and sales stakeholders	Validation — confirming the product meets user needs
Testing pyramid value	Integration-level bugs isolated faster than if first discovered in acceptance; lower debugging cost	Testing efficiency — earlier detection = cheaper resolution

Source: Assignment Activity Unit 7 — MSIT 5910-01 Capstone Project, University of the People, Term 4 AY2026.